Name	Shiven Saxena
Student ID	34997407
Studio Session	17

Part A: URL of the Visualisation

Paste your visualisation URL in the empty space below.
Tip: Test your URL in incognito mode to ensure it is a working URL

https://ssax0009-ctrl.github.io/DV2_FIT3179/

Part B: Domain Description

Describe your domain chosen for DV II

- The domain for this visualisation is: Australian women's healthcare, and how hospital care for hysterectomy and endometriosis is affected by state, local hospital network, funding sector and age.
- The users are health-service planners and women's health advocates, who understand complex terms such as median wait, LHN, SA3, age-standardised rate, SES group and suppressed data.
- Their workflow is constrained by public-only waiting times, waits and rates from different years, and withheld values that must not be read as short waits.

Describe the Why and Who aspects of your DV II

Why
The goal is to survey how affordable and accessible Women's health is through looking at the statistics of the two of the most common reasons for hospitalisations Endometriosis and Hysterectomy.

Who (in the perspective of Australians):

- Health Service Planners and Policy makers:
    - We analyse to the local level and show how this differs from the state level, integral for local governments to know how to "plan ahead"
- Women Health advocates:
    - Showing trends and how accessibility is improving over time, especially when different groups are in perspective
    -
- General Women (especially those with Endometriosis or Hysterectomy):
    - Analyse the how accessibility for treatment and where is ready for them

Describe the "What" aspect of your DV II

Not I have gotten CSV files from other CSV files which is under author, when this is the case AIHW is the author

Data name	**Author**	**Use**	**Why?**
Hysterectomy hospitalisations, 2014–15 to 2021–22	AIHW Australian Atlas of Healthcare Variation (National Hospital Morbidity Database, ABS population estimates). Women 15+, age-standardised to the 2001 population.	State and SA3 rates, Indigenous status, remoteness and SES, private share (Figures 3–11)	Hysterectomy is a very common surgery and can be described as a benchmark for all surgeries, this data is comprehensive on locations, as well as sectors.
Elective surgery waiting times, 2024–25	AIHW *Australian hospital statistics*, Table 4.7 for state medians, plus LHN files with coordinates	Figures 1–2	Observing wait times shows us how accessible a procedure is
Endometriosis hospital procedures (PHE 357)	AIHW, 2026 supplementary tables	Admissions, urgency, age, sector (Figures 12–15)	Most of the data for endometriosis which is one of the most prevalent illness specific to women
Boundaries	ABS ASGS Edition 3 (SA3); geoBoundaries gbOpen (state outlines)	Maps	For making maps

Hospital IDs, names and coordinates	Hospital IDs, names and coordinates	Hospital IDs, names and coordinates	Hospital IDs, names and coordinates used for other graphs

Some notes on the Creation Process:

- Missing Data was often an issue, especially when 2 graphs are complementary, so I made the difficult decision to suppress both in order to prevent misleading information
- Some data was not all in the same place, such as coordinates, so I had to find that separately.
- One of the biggest problems was quality data, as there were not many actual data providers. Weighing the odds, I decided to use old data 2021 - 2022 that is credible over non credible data that is relevant.
- Figure 1 and 2

Describe the "How" aspect of your DV II

Figure	Idiom	Reasoning
1	Choropleth map	Help show density of state, and can be used to compare for outliers (figure 2)
2	Proportional symbol map	Shows the movement of densities (closer to the coast the more hospitals for example) help compare with figure 1
3	Ridgeline graph	Shows the distribution of women for hospitalisations, allows us to see how location affects the average for each state (exposing lying with state averages)
4	Interval plot	Similar to ridgeline, but further shows exemplifies the idea of the outliers.
5	Dot Map	Shows that variation is spread geographically and what regions are most sparse

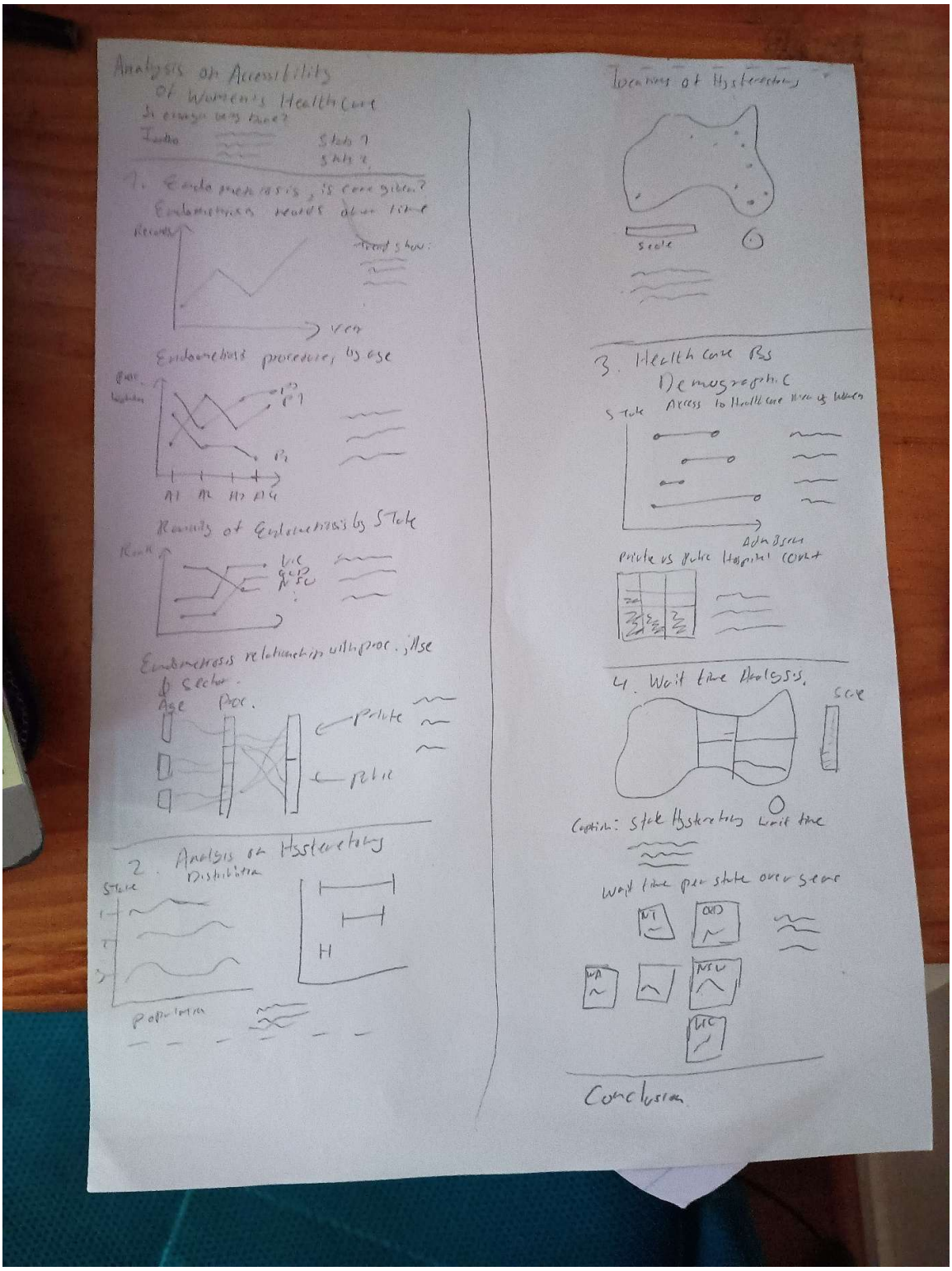
6	Radial chart	Combines two factors at once: remoteness by colour and SES by slice. It demonstrates that the highest rates are not in the most remote areas.
7	Dumbbell	Two groups (First nation and Others) per state with the gap drawn, so the direction of the difference is the first thing seen. Helps us compare if First Nation people are getting the care they deserve
8	Waffle	One square equals 1%, so private share is counted, not estimated. An easy visual to see how private hospitals dominate over public hospitals.
9	Slope chart	Compares 2 main years, allows us to see whether the surgery admissions increase or decrease.
10	Tile-grid map with sparklines	Equal-size tiles stop Tasmania, the ACT and the NT from vanishing on a real map. Sparklines add the shape between the endpoints. A fun intuitive map that shows rates per state.
11	Heatmap	Another usage of distribution, but here we compare state and time. Allows us to see a projection of how hysterectomy records are falling
12	Time series	Shows growth/decline of endometriosis rate.
13	waterfall	Demonstrates ideas such as medical urgency as well as assisting the time series
14	Parallel Plot	Ten procedures are muted for context. Two (laparoscopic excision and cystoscopy) are emphasised to show different age peaks and how age plays a role on surgery given and accessibility.

Special Features:
Used a button for glossary for ease of use.

AI Declaration (also on visualisation):
AI was used in this assignment with LLMs such as Claude, Copilot, Chatgpt and Gemini to help debug code, ease of code format and structuring as well as giving advice.

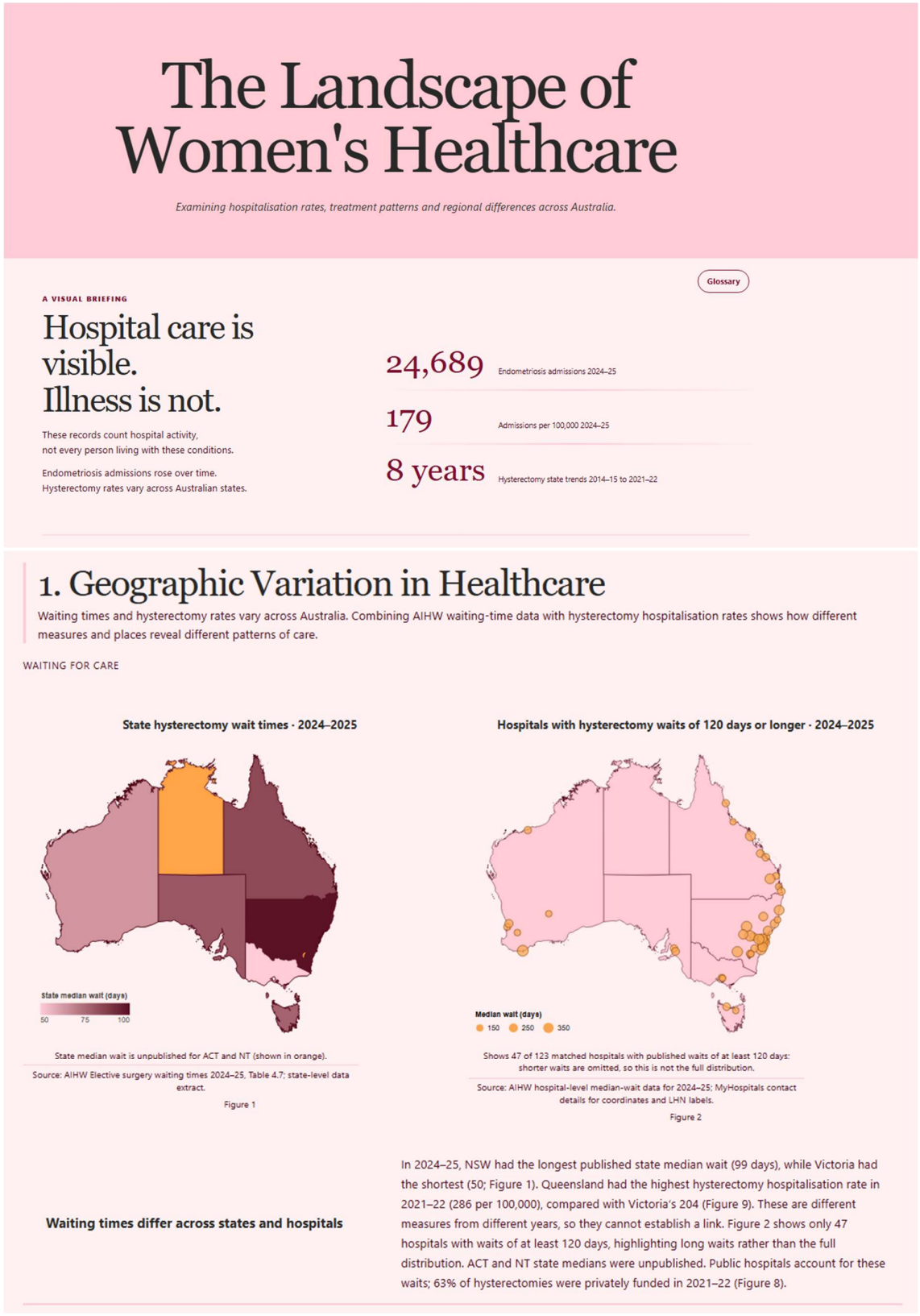
Part C: Sketch

https://github.com/ssax0009-ctrl/DV2_FIT3179/blob/main/sketch.pdf



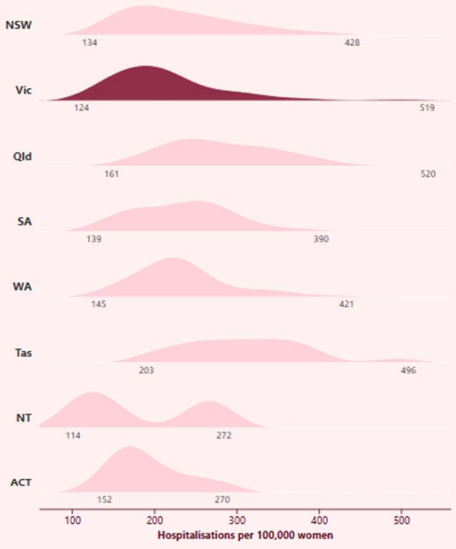
Part D: Full-size screenshot of your DV II

Paste a full-size screenshot of your DV II assignment from the URL here



VARIATION WITHIN STATES

SA3 hysterectomy hospitalisation rates by state · 2021–2022

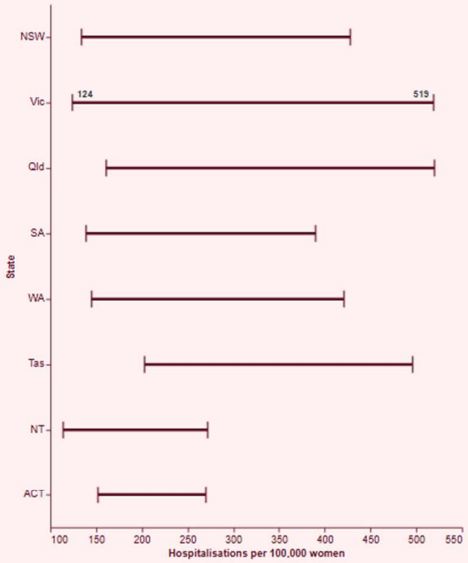


Each ridge smooths the SA3 rates in one state; numbers mark its lowest and highest area. Suppressed rates are omitted, and the ACT and NT curves rest on only 5 to 6 areas.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2021–22.

Figure 3

SA3 hysterectomy hospitalisation rate ranges by state · 2021–2022



24 SA3 values are suppressed; statewide counts of valid SA3s range from 5 to 87.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2021–22.

Figure 4

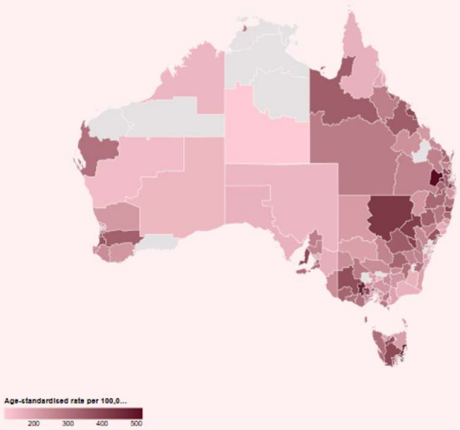
State averages conceal substantial local variation

Figures 3 and 4 show that areas within the same state can have very different hysterectomy hospitalisation rates. In Victoria, the range runs from 124 per 100,000 women in Whitehorse–West to 519 in Ballarat. The key story is not simply that states differ, but that the place someone lives within a state may also matter.

Local differences become visible on the map

Figure 5 maps the published SA3 rates onto their actual 2021 ABS statistical-area boundaries, adding geographic context to the ranges in Figure 4. Darker shading indicates higher age-standardised hospitalisation rates; grey areas have suppressed rates. The map suggests some broad geographic patterning, but rates also vary sharply between nearby areas, so the overall picture is far from uniform. The interval plot makes the lowest-to-highest values easier to compare.

Age-standardised hysterectomy hospitalisation rates by SA3 · 2021–2022



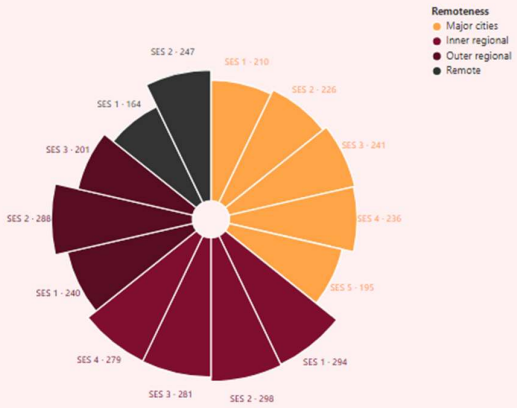
316 published SA3 rates are mapped to 2021 ABS boundaries; 20 suppressed state and territory rates are shaded grey. Other Territories are excluded.

Source: AIHW hysterectomy workbook, Hospitalisations (SA3), 2021–22; ABS ASGS Edition 3 SA3 boundaries, GDA2020.

Figure 5

REMOTENESS AND SOCIOECONOMIC CONTEXT

Hysterectomy hospitalisation rates by remoteness and SES group · 2021–2022



The highest rates are not found in the most remote areas

In Figure 6, inner regional areas record 292 hospitalisations per 100,000 women, above major cities at 218 and remote areas at 193. This challenges a simple assumption that remoteness alone explains differences in care. The socioeconomic slices also show variation within remoteness categories, suggesting that geography and social circumstances should be considered together rather than as competing explanations.

Area: age-standardised rate/100k; colour: remoteness; slices: SES; Outer Regional/Remote SES suppressed.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2021–22.

Figure 6

2. Healthcare Patterns by Demographics

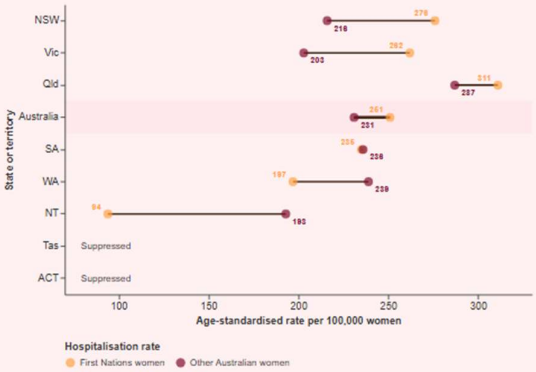
Differences in rates, waiting times and funding show that healthcare experiences are not uniform. Comparing these measures helps identify where further questions about access and service delivery arise.

FIRST NATIONS WOMEN AND HYSTERECTOMY RATES

National averages hide different state patterns

In 2021–22, First Nations women had a higher national hysterectomy hospitalisation rate than Other Australian women (251 versus 231 per 100,000). However, Figure 7 shows contrasting state-level patterns: First Nations women had higher rates in NSW, Victoria and Queensland, while Other Australian women had higher rates in WA and the NT. These results highlight how national averages can conceal geographic variation in recorded hysterectomy hospitalisations.

Hysterectomy hospitalisation rates by Indigenous status and state · 2021–2022



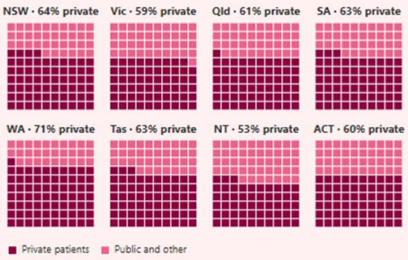
Hospitalisation rates are not measures of prevalence or need; Tasmania and ACT are suppressed.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2021–22.

Figure 7

PUBLIC AND PRIVATE FUNDING

Private-patient share of hysterectomy hospitalisations by state · 2021–2022



Each square is 1% of a state's hysterectomy hospitalisations; nationally, 63% were private.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2021–22.

Figure 8

Private hospitals account for most hysterectomy admissions

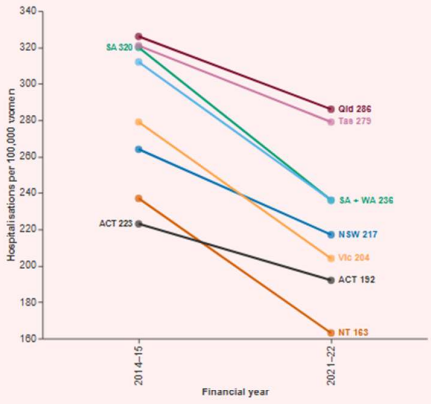
Figure 8 shows that 63% of hysterectomy hospitalisations were privately funded nationally. The share ranges from 53% in the NT to 71% in WA, making the private sector the larger part of recorded activity in every state shown. This uneven funding mix matters because national totals can make the system look more uniform than the state-level picture suggests.

3. Temporal Trends in Hysterectomy Rates

Hysterectomy rates generally declined over the period, but the pace and shape of change differed by state. Looking at the full trajectory reveals more than a single national trend.

CHANGE OVER TIME

Hysterectomy hospitalisation rates by state · 2014–2022

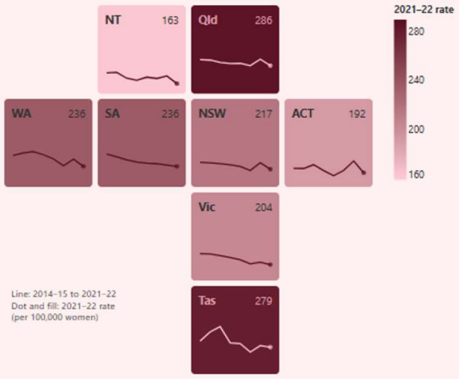


Age-standardised rates; SA and WA finish at the same rate.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2014–15 and 2021–22.

Figure 9

Hysterectomy hospitalisation rates by state · 2014–2022



Fill shows each state's 2021–22 age-standardised rate per 100,000 women; each line runs from 2014–15 to 2021–22.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2014–15 to 2021–22.

Figure 10

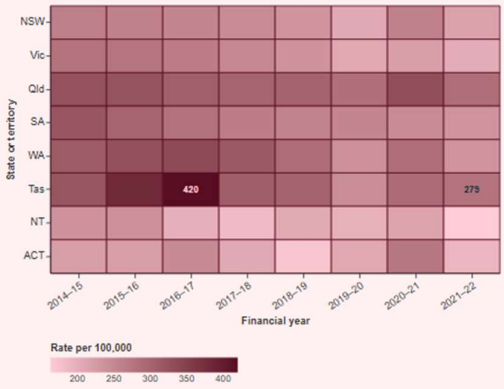
Declining rates did not erase differences between states

Figure 9 shows broad declines in hysterectomy hospitalisation rates, while Figure 10 shows how trajectories differed between states. Comparing only the 2014–15 and 2021–22 endpoints, the gap between the highest and lowest rates widened from 103 to 123 per 100,000 women. The gap was wider in between: in 2016–17, Tasmania's rate of 420 versus the NT's 200 created a 220-point difference (Figure 11). The endpoints alone therefore miss important variation over time.

Tasmania followed a different trajectory

Figure 11 highlights Tasmania's rise to 420 hospitalisations per 100,000 women in 2016–17, followed by a fall to 279 in 2021–22. That peak stands out against the broader pattern of declining rates and shows why a start-to-end comparison can miss important changes in between. The trajectory is unusual, but the data alone do not identify its cause.

Annual age-standardised hysterectomy hospitalisation rates by state · 2014–2022



Changes around 2019–20 to 2021–22 overlap with COVID-era restrictions.

Source: AIHW Australian Atlas of Healthcare Variation Time Series, 2014–15 to 2021–22.

Figure 11

4. Endometriosis Hospitalisations and Procedures

Endometriosis hospitalisations have increased, while procedure patterns differ by age and hospital sector. Together, these charts show how recorded treatment is distributed and not the full prevalence of the condition.

GROWTH IN HOSPITAL ACTIVITY

Annual endometriosis admissions and crude admission rates · 2015–2025

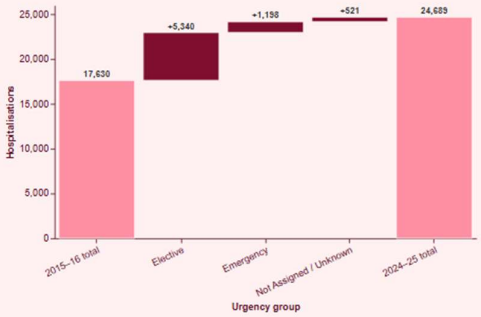


Crude rates are not age-adjusted and are not a measure of disease prevalence.

Source: AIHW PHE 357, Endometriosis hospital procedures, 2015–16 to 2024–25.

Figure 12

Change in endometriosis admissions by urgency · 2015–2025



The waterfall compares total admissions at the start and end of the series.

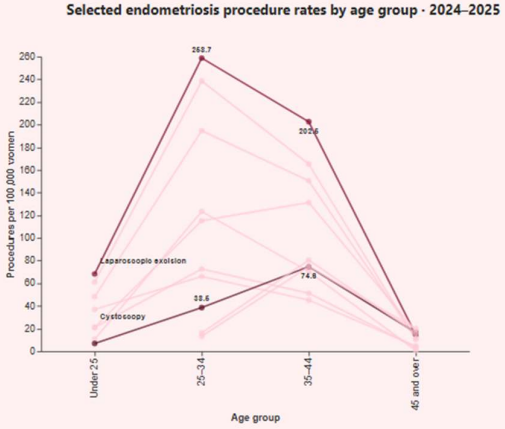
Source: AIHW PHE 357, Endometriosis hospital procedures, 2015–16 and 2024–25.

Figure 13

The rise in endometriosis admissions was driven mainly by elective care

Figure 12 shows admissions rising from 17,630 to 24,689 between 2015–16 and 2024–25, an increase of about 40%. Figure 13 indicates that elective admissions account for around 76% of that growth. The increase is therefore concentrated in planned hospital care, making elective treatment an important part of the story behind rising recorded activity.

AGE AND PROCEDURE TYPE



Rates are for selected procedures in 2024–25, not unique people.

Source: AIHW PHE 357, Endometriosis hospital procedures, 2024–25.

Figure 14

Procedure patterns peak during the main reproductive years

In Figure 14, six of ten selected procedures peak at ages 25–34 and four at ages 35–44. This pattern is consistent with endometriosis care being concentrated during reproductive years. Symptoms often improve after menopause, although the condition can persist. The decline in older age groups should therefore be read as a pattern in recorded procedures, not evidence that endometriosis disappears after a particular age.

5. Key Findings and Implications

The data reveal variation in care, but not its full explanation

Figures 1–14 show variation in waiting times, hysterectomy hospitalisations, funding and endometriosis procedures. The highest hysterectomy rate and longest state wait occur in different states, but these measures cover different years and cannot be linked causally. Rates also differ by Indigenous status and funding sector. Endometriosis admissions increased, mainly through elective care, while recorded procedures peak in reproductive-age groups. These patterns describe hospital activity, not the full level of need or access. Further evidence is needed to explain why care differs by place and population.

TERMS TO KNOW

A short glossary.

AIHW
Australian Institute of Health and Welfare, the national agency that publishes health and welfare statistics.

Age-standardised rate
A rate adjusted to account for differences in age between populations, making comparisons fairer.

Crude rate
The number of events relative to a population, without adjusting for its age profile.

Endometriosis
A condition where tissue similar to the lining of the uterus grows elsewhere in the body.

Hysterectomy
An operation to remove the uterus.

LHN
Local Hospital Network: a group of public hospitals and health services serving a region.

Median wait time
The middle wait time: half of the patients waited less, and half waited longer.

SA3
Statistical Area Level 3, a geographic area used by the Australian Bureau of Statistics to report regional data.

SES
Socioeconomic status: a way of describing social and economic circumstances, such as relative advantage or disadvantage.

Suppressed data
Figures not published by the data source, often to protect privacy or because estimates are unreliable.

REFERENCES

Data sources (APA 7th edition).

Australian Bureau of Statistics. (2021). *Australian Statistical Geography Standard (ASGS): Edition 3, July 2021–June 2026—Digital boundary files* [Data set]. Download the 2021 SA3 boundary file (GDA2020). Figure 5 uses the 2021 SA3 boundaries in GDA2020, joined to AIHW records by SA3 code.

Australian Institute of Health and Welfare. (n.d.). *Australian Atlas of Healthcare Variation time series 2014–15 to 2021–22: Hysterectomy hospitalisations* [Data set]. Project workbook (Excel). Worksheets used include Hospitalisations (SA3), Hospitalisations (State), Remoteness x SES, Indigenous status (State), and Patient funding status (State). Source notes identify the National Hospital Morbidity Database and ABS estimated resident populations.

Australian Institute of Health and Welfare. (n.d.). *Elective surgery waiting times 2024–25: Australian hospital statistics* [Data set]. AIHW data workbook (Excel copy). Table 4.7 is the source for state-level hysterectomy waiting times.

Australian Institute of Health and Welfare. (2026). *Endometriosis hospital procedures: Supplementary data tables* (Cat. no. PHE 357) [Data set]. Project workbook (Excel). Tables 1, 2 and 4 provide the time-series, urgency and age-group data.

geoBoundaries. (2023). *geoBoundaries: Australia, administrative level 1 (gbOpen)* [Data set]. https://www.geoboundaries.org/api/current/gbOpen/AUS/ADM1/.

Australian Institute of Health and Welfare. (2025, November 25). *Waiting times by intended procedure*. https://www.aihw.gov.au/hospitals/topics/elective-surgery/waiting-times-by-surgical-specialty-waiting-times-by-intended-procedure-a57ccc30427bd0cc3ab50bc99d244ed3. Figure 2 uses the 2024–25 hospital-level hysterectomy values from this AIHW source; blank and unpublished values are not mapped.

Australian Institute of Health and Welfare. (n.d.). *Hysterectomy waiting times by state, 2024–25* [Data set]. State data extract (CSV). State figures are from Table 4.7 of *Elective surgery waiting times 2024–25: Australian hospital statistics*.

Jorm, L., & Barbieri, S. (2019). *Hospital IDs, names and coordinates (CSV)* [Data set]. figshare. https://doi.org/10.6084/m9.figshare.8319737.v1. Coordinates were used to locate the hospital-level waiting-time records; LHN labels and bed-size categories are from the supplied MyHospitals contact-details CSV.

Runfola, D., Anderson, A., Baier, H., Crittenden, M., Dowker, E., Fuhrig, S., Goodman, S., Grimsley, G., Layko, R., Melville, G., Mulder, M., Panganiban, J., Peck, A., Seitz, L., Shea, S., Slevin, H., Youngerman, R., & Hobbs, L. (2020). geoBoundaries: A global database of political administrative boundaries. *PLOS ONE, 15*(4), Article e0231866. https://doi.org/10.1371/journal.pone.0231866.

Hysterectomy rates cover women aged 15 and over and are age-standardised to the 2001 Australian population. Waiting-time measures are days waited by admitted patients; state percentiles are unpublished for the ACT and NT. Endometriosis rates are crude rates reported by AIHW. The ACT did not provide private-hospital data for 2024–25.

Visualisation by Shiven Saxena · 2026 · AI assistance supported code, chart specifications and editing. The analysis combines AIHW hysterectomy hospitalisation rates through 2021–22 with AIHW elective-surgery waiting times for 2024–25. These datasets measure different outcomes in different years.